

# Thorben Foerst

DevOps & Cloud Enthusiast

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## Summary

IT Management student (B.Sc.) at the Technical University of Cologne, specialising in cloud and DevOps topics. AWS Certified Solutions Architect – Associate, with practical experience in setting up Kubernetes clusters using Terraform and working with GitOps workflows via ArgoCD.

## Skills

### Cloud (AWS)

VPC, IAM, EC2, S3, RDS, EKS, Secrets Manager and many more (SAA-C03-certified)

### Containers & Orchestration

Kubernetes, k3s, Amazon EKS, Docker, Docker Compose, Helm

### IaC & CI/CD

Terraform, GitHub Actions, ArgoCD (GitOps), Ansible, Git

### Monitoring & Observability

Prometheus, Grafana, Uptime Kuma

### Network & Infrastructure

VLANs & subnetting, DHCP/DNS/firewall configuration, Linux, TrueNAS, reverse proxies (Traefik, Nginx Proxy Manager), load balancing, WireGuard

### Programming

Python, HCL, Bash, YAML, JSON, SQL, C#, C++

### AI

RAG, LangChain, Chroma, OpenAI API

## Certifications

### AWS Certified Solutions Architect – Associate

SAA-C03 – Expires: 6 May 2029  
Credly verification

## Languages

German (native language)  
English (C1)

## Education

### B.Sc. in IT Management

TH Köln – Campus Gummersbach

Oct. 2022 – expected Feb. 2027

- Bachelor's thesis: *"GitOps-based vs. pipeline-based deployments in Kubernetes"*

## Experience

### Software developer

Foerst GmbH

Oct. 2021 – Sept. 2022 & Oct. 2025 – March 2026

- Design and development of interactive training modules for driving simulation software
- Development of an in-house RAG-based chatbot (Streamlit, OpenAI API, ChromaDB) with a knowledge base of 330 pages, used by the development department to assist with the creation of new exercise modules and, according to feedback from the team, saving time in day-to-day development work
- Containerisation and deployment of applications on on-premises servers using Docker and Docker Compose
- Induction and professional support for new staff

## Projects

### Home Lab

Personal project

June 2025 – ongoing

- Setting up a custom-designed network with its own routing and VLAN segmentation as the basis for the entire infrastructure
- Virtualisation layer on two Proxmox nodes with 6 VMs; running a Kubernetes cluster (k3s) on top of this, comprising 2 master and 2 worker nodes to ensure resilience against the failure of individual nodes
- GitOps deployment via ArgoCD for 11 services in production (including Nextcloud, Immich and Vaultwarden), Longhorn storage with automated S3 backups
- Infrastructure-as-Code using Terraform, and monitoring with Prometheus and Grafana

### Practical project work

TH Cologne, module in the final semester, forming the basis of the Bachelor's thesis

Apr. 2026 – Aug. 2026

- Setting up a production-ready Kubernetes infrastructure in the AWS Cloud using Terraform, including the deployment of a managed Kubernetes cluster (Amazon EKS)
- Containerisation and deployment of a test application with a database connection (Amazon RDS) for end-to-end validation of the infrastructure
- Secure management of database access credentials via AWS Secrets Manager, and provisioning via the AWS Load Balancer Controller (LBC)